

Rangarajan G

Agentic AI Engineer | GenAI Engineer | LLM Orchestration

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PROFESSIONAL SUMMARY

Generative AI Engineer specializing in agentic AI systems, LLM orchestration, and Retrieval-Augmented Generation (RAG) architectures. Experienced in building production-grade multi-agent pipelines using Google ADK, LangGraph, and MCP tooling including a CLI-based agentic PLM assistant enabling LLM-driven tool selection, RAG-based document retrieval, and multi-turn conversational reasoning across complex PLM workflows. Hands-on expertise in LoRA fine-tuning, prompt engineering, vector databases, and self-healing AI automation. Delivered measurable impact including 70% reduction in flaky tests, 40% reduction in manual design effort, and 30% improvement in ticket resolution time.

TECHNICAL SKILLS

Agentic AI and LLM Orchestration: Google ADK, LangGraph, LangChain, Crew AI, MCP, Multi-Agent Pipelines, Tool Calling, RAG, Prompt Engineering

LLMs and AI Models: Gemini, LLaMA, LoRA Fine-Tuning, Embeddings, Vector Databases (Pinecone, Chroma, FAISS), Claude Code, Codex, Copilot.

AI Automation and Testing: Playwright, Self-Healing Frameworks, AI-driven UI and API testing.

Software Engineering: Python, Typescript, FastAPI, Flask, REST APIs, PostgreSQL, Git, Agile

Cloud and Platforms: AWS Bedrock, AWS SageMaker, Docker, Jira API, MCP Integrations

Enterprise Technologies: Centric PLM, Product Lifecycle Management, Configuration Management, Enterprise Solution Delivery, Customer Consulting, Migration Support, Production Deployment, UAT, Go-Live Support.

PROFESSIONAL EXPERIENCE

AI Automation Engineer | Kripya Solution Private Ltd | Chennai, India

Apr 2025 - Present

- Own AI-powered MCP tooling for customer-specific Centric PLM engineering workflows, supporting enterprise customers across US and Europe
- Led design of a Google ADK multi-agent system for autonomous test case generation, cutting manual design effort by 40% and story-to-test-case turnaround by 50% across 30+ sprint cycles.
- Built a self-healing Playwright + Gemini automation framework that recovers locators from DOM snapshots, reducing flaky tests by 70% and improving CI reliability.
- Introduced structured agent personas with reasoning constraints and output validation, reducing hallucinations in structured reporting.
- Delivered a CLI-based agentic Centric PLM assistant (LangGraph + Gemini) for tool selection, RAG retrieval, and code/config generation across complex PLM workflows.
- Authored 12+ reusable Claude-powered prompt skills for BDD generation, validation logic, and migration support, adopted across enterprise PLM operations.

AI/ML Engineer Intern | Tristha Global Pvt Ltd | Chennai, India

Feb 2025 - Apr 2025

- Fine-tuned LLaMA 3.2-11B with LoRA for banking-domain classification, improving accuracy by 25% and cutting false positives by 35%.
- Built a RAG pipeline with vector-store retrieval, reducing hallucinations and improving domain-specific answer precision.

Python Developer | Office 2000 Solutions Pvt Ltd | Chennai, India

Jul 2024 - Nov 2024

- Optimized backend REST APIs and PostgreSQL queries, cutting response latency by 50%.
- Built a real-time power metrics dashboard (Flask) monitoring 10+ live data streams.

KEY PROJECTS

agentkit-mcp | MCP, Python, Gemini, OpenRouter <https://github.com/rangarajan19/agentkit-mcp>

- Built an MCP-first agent framework enforcing tool-layer guardrails, including dry-run mode, human approval, argument rules, and tool allow-lists, to constrain agent actions before execution
- Designed an agent loop with tracing and step limits where tool errors route back to the model for recovery, validated with 47 automated tests in CI.
- Supported Gemini and free OpenRouter models with automatic retry and model fallback; shipped example agents for GitHub-integrated issue triage and web research.

Agentic Api Testing | LangGraph, Ollama, Gemini, Python <https://github.com/rangarajan19/agentic-api-testing>

- Built a chat-driven API testing tool where a local LLM agent (LangGraph + Ollama) interprets and routes natural-language test requests, while deterministic pytest assertions perform the actual verification, keeping LLM steering separate from test reliability.

BDD Feature Architect | Claude API, Prompt Engineering, Python

- Automated Behavior-Driven Development (BDD) Feature File generation by converting structured test cases into Gherkin scenarios using Claude API and a template-driven prompt workflow.
- Automated 200+ Gherkin scenarios, cutting manual BDD authoring time by **50%** while ensuring full traceability to source test cases.

Sign Language Caption Generation | LSTM, MediaPipe, Python

- Designed an LSTM-based sign-language recognition model achieving **60% higher accuracy** than the baseline using MediaPipe for real-time hand landmark detection.
- Research published in **IEEE Xplore (2024)**, demonstrating potential for real-world accessibility applications in assistive communication technology.

EDUCATION

Bachelor of Technology in Information Technology

2020 - 2024

Rajalakshmi Engineering College, Chennai | CGPA: 8.39 / 10

CERTIFICATIONS AND PUBLICATIONS

Publication

- "Sign Language Caption Generation Using LSTM" - IEEE Xplore, 2024

Certifications

- **AWS Certified AI Practitioner** - Amazon Web Services (AWS)
- Prompt Engineering for Developers - Great Lakes Institute of Management
- Essential Automation Professional - Automation Anywhere University
- UiPath Studio Foundational - UiPath

OPEN SOURCE CONTRIBUTIONS

HelpCode-ai/anythingmcp — github.com/HelpCode-ai/anythingmcp - Added an **adapter:new** scaffolder so new connector adapters can be generated from a template rather than built from scratch (PR #698, closes #585, merged).

agentevals-dev/agentevals — github.com/agentevals-dev/agentevals - Fixed a CI gap where autofixable Ruff lint violations were passing silently instead of failing the build, tightening code-quality enforcement for the eval framework (PR #224, merged).